



BrightLogic ActiveFloor Content Development

Reading Sense Data from the ActiveFloor

To read sense data from the ActiveFloor, your application will need to make HTTP GET requests to the ActiveFloor Server (up to 30 times per second). The application that provides the sense data is the “BLFloorServer.exe” which is running on the BrightLogic server, this application is responsible for constantly knowing the state of the sensors on the ActiveFloor. When an HTTP GET request is made to the BLFloorServer e.g. <http://192.168.1.1:8080>, the server will respond with an XML response, similar to that shown below:

```
<?xml version="1.0" encoding="UTF-8"?>
<BLFloor ledsX="192" ledsY="192" sensorsX="24" sensorsY="24">
<Rows date="2016-08-24 11:59:35.280">
<Row rownum="0" values="....." />
<Row rownum="1" values="....." />
<Row rownum="2" values="....." />
<Row rownum="3" values="....." />
<Row rownum="4" values="....." />
<Row rownum="5" values=".....*" />
<Row rownum="6" values=".....*,*" />
<Row rownum="7" values="....." />
<Row rownum="8" values="....." />
<Row rownum="9" values="....." />
<Row rownum="10" values="....." />
<Row rownum="11" values=".....*,*" />
<Row rownum="12" values=".....*" />
<Row rownum="13" values="....." />
<Row rownum="14" values="....." />
<Row rownum="15" values="....." />
<Row rownum="16" values="....." />
<Row rownum="17" values="....." />
<Row rownum="18" values="....." />
<Row rownum="19" values="....." />
<Row rownum="20" values="....." />
<Row rownum="21" values="....." />
<Row rownum="22" values="....." />
<Row rownum="23" values="....." />
</Rows>
</BLFloor>
```

Each <Row> element represents all of the sensors in that row of ActiveFloor panels on the floor. E.g. in the example above, we have a 3 panel x 3 panel layout (a total of 24 sensors x 24 sensors). Each panel has a grid of 8x8 sensors, therefore showing 24 sensors per <Row> element (shown in the attribute field). In the values attribute, each sensor is represented by either a “.” or a “*”, the “.” Indicates that the sensor is currently not blocked by an object/foot/hand etc, the “*” indicates that the sensor is blocked at the time the HTTP GET Request was made. As you can see from the values listed above the attribute values are comma delimited.





In the <Rows> Element, there are 24 Child <Row> nodes, each one representing a row of sensors on the interactive floor. With a 3 Panel x 3 Panel setup, there is a total of 24 x 24 sensors = 576 Sensors.

Refreshing the ActiveFloor Data

The sensor data on the BrightLogic ActiveFloor Server is refreshed approximately 30 times per second, with an approximate delay of 33 milliseconds from touch to visual reaction, therefore its important to constantly load new sense data into your application at a rate of 30 times per second.

A simple javascript application showing the raw sense data on the ActiveFloor. This application requires a the <canvas> HTML TAG with an element id of “floorCanvas”. The following example also requires the latest version of JQuery.

```
function initCanvas(arr) {  
    'use strict';  
  
    canvas = document.getElementById('floorCanvas');  
  
    canvas.width = ledsX;  
    canvas.height = ledsY;  
  
    context2D = canvas.getContext('2d');  
  
    var i, tempRow, p, srchStr, tempX, tempY;  
    for (i = 0; i < arr.length; i += 1) {  
        tempRow = arr[i];  
  
        for (p = 0; p < tempRow.length; p += 1) {  
            srchStr = tempRow.substring(p, p + 1);  
            if (srchStr === charSearch) {  
                tempX = p * ledPerSensorX;  
                tempY = i * ledPerSensorY;  
  
                drawObj('square', tempX, tempY, 5);  
            }  
        }  
    }  
}
```





```
    }  
}  
  
function refreshXML() {  
    'use strict';  
  
    $.get('http://127.0.0.1:8080/', function (data) {  
        dataHolderArray = [];  
  
        $(data).find('BLFloor').each(function () {  
            $item = $(this);  
  
            ledsX = $item.attr('ledsX');  
            ledsY = $item.attr('ledsY');  
            sensorsX = $item.attr('sensorsX');  
            sensorsY = $item.attr('sensorsY');  
            ledPerSensorX = (ledsX / sensorsX);  
            ledPerSensorY = (ledsY / sensorsY);  
            xCenter = ledPerSensorX / 2;  
            yCenter = ledPerSensorY / 2;  
        });  
  
        $(data).find('Row').each(function () {  
            var $row, rowNum, rowVal, n;  
  
            $row = $(this);  
            rowNum = $row.attr('rownum');  
            rowVal = $row.attr('values');  
            n = rowVal.split(charDivide).join('');
```





```
        dataHolderArray.push(n);  
    });  
  
    initCanvas(dataHolderArray);  
});  
}
```

